

UNIVERSITY OF
Waterloo



Department of Mechanical and Mechatronics Engineering

**The Design of a Reefing Parachute
Recovery System for a Student
Competition Sounding Rocket**

A Report Prepared For:
The University of Waterloo

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Prof Michael Collins
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Dear Professor Collins,

This report, entitled *The Design of a Reefing Parachute Recovery System for a Student Competition Sounding Rocket*, was written to fulfil my 2B work term report requirements. This is my 2nd work report submission.

In this project, a rocket recovery system was designed. The rocket in question is ‘Kraken of the Sky’, a high-altitude sounding rocket built by Waterloo Rocketry design team, where I am a senior member of the recovery sub-team. The previous system had suffered issues, and was seen as a definite area of improvement. It is for this reason that myself and the other members of the recovery sub-team pursued this project. The engineering work carried out in the process of developing this new system will be described in detail throughout the report.

This report was written entirely by me, and has not received any previous academic credit at this or any other institution. I would like to acknowledge the help of Christopher Sankey and Roman Kobets, the respective current and former heads of the recovery sub-team on the Waterloo Rocketry design team. They performed a large portion of the development work outlined in the report with me and provided some assistance in the preparation of this document. The university provided a large number of resources for the design team and their work in general. The sponsors of Waterloo Rocketry have also provided a lot of parts and money, which allowed the team to pursue this project in the first place. I would also like to specifically thank the Region of Waterloo, GL Heli, Liftwerx, and Prairie Twister for assistance in the specific work that will be covered in this report.

Best Regards

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Summary

In this project, a parachute recovery system was constructed for a rocket that is to participate in a student competition. The overall goals of the rocket system are broadly defined as part of the background, with the main goal being victory in the competition. A problematic recovery system from the previous year was analyzed from an engineering perspective, and several parts of it were selected to undergo a re-design to boost reliability and other performance characteristics of the system. The specific parts of the system that were re-designed were the line cutter system and the parachutes themselves.

Upon an engineering analysis of several designs for both systems, directions for development were defined. The cutters had previously been pyrotechnic cutters, but the new design called for the development of a cheaper and simpler pyrogenic epoxy line burning system. However, development into the system revealed insurmountable cutting delay problems.

The parachutes had previously consisted of a drogue parachute and a main parachute to satisfy specific competition descent speed requirements. The new system selected for development was a single reefed parachute. The exact physical characteristics of this system were identified.

Several different designs for this parachute were selected. This was narrowed down to two for testing. A number of different testing strategies were then considered for testing the parachute, before being narrowed down to truck pull testing with the use of a custom rig and a smaller scale parachute. Initial riggings plans were based on geometric calculations using the constructed size of the parachute.

After difficulties with testing, plans were changed around. The full scale parachute was used in place of the small scale one, and a modification to the rigging was proposed. It became apparent that geometric calculations would be wildly inaccurate, and extensive testing to accurately determine the correct lengths would be required. While this could not be completed initially, the rigging modification proved to be successful.

The requirements were changed after some administrative developments. Access to new pyrotechnic cutters and a chartered helicopter, as well as a change in competition requirements allowed for a change in the design plan. Various preparations were done to execute a helicopter drop test, but it could not be carried out due to regulatory issues.

1.0 Introduction

Waterloo Rocketry is a student team at the university of Waterloo that designs rockets for the Spaceport America cup, a yearly competition [1]. Every year, a new sounding rocket is built to compete in one of the competition categories. A sounding rocket is a rocket that is meant for atmospheric flight, and is not designed to reach space. However, the design of sounding and orbital rockets have many things in common.

For the 2020 rocket design, the '30K SRAD' category was chosen to compete in by the team. For this category, the rocket must attain an apogee (the highest point of the flight) of as close to 30 000 feet AGL (above ground level) as possible. Moreover, the engine for a rocket in this category must also be designed and built by students. Additional points are awarded for successful vehicle recovery and technical competence [2].

The sounding rockets that the team build are fairly standard in their architecture. The rocket possesses an engine to propel it, along with fuel and oxidizer to run it. A suite of electronics including control all of the rocket functions. The airframe of the rocket keeps all of the internal components on an aerodynamic shape that can sustain stable flight. The rocket also contains a payload (usually a scientific experiment), and a parachute descent system that allows the recovery of the vehicle. This last specific part, the recovery system, will be the main subject of this report. A diagram of all these parts can be seen below at the end of the section.

The broad aim of this large system as outlined by the team is to win the competition, and to also provide students the ability to work on interesting technical projects. This latter point is relevant since competition rewards technically challenging, creative or novel ideas in the design. While not a concrete design requirement, this latter point was considered when making design decisions. The stated design goal for the rocket was to achieve the correct height and successfully land. It must be developed and readied for flight within one year. However, due to the Covid-19 pandemic, this timeline ended up being extended to two years.

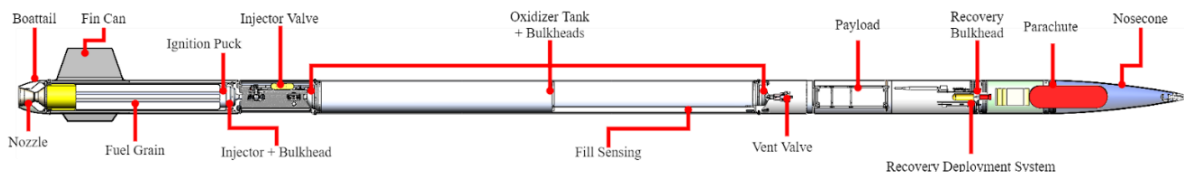


Figure 1: Overview of the rocket system [3]

2.0 Recovery System & Main Problem Statement

The parachute system of the rocket is a flight-critical system that is required for a successful recovery. However, in these the previous competition cycles, this system failed to work correctly several times. Due to this, it was subject to a substantial re-evaluation and redesign for the 2020 competition cycle.

A significant limitation is imposed on the system by the competition rules. To increase rocket turnaround and mitigate the risk of drift outside of bounds, the rules state that the recovery system must have two stages. From the apogee and down to 1500 feet AGL, the rocket must descend at a speed of 75-150 feet per second. Only when it is below 1500 feet, the deployment of the main parachute can take place, with a recommended final descent speed of less than 30 feet per second [1].

The system was controlled by a flight computer. The flight computer was connected to two altimeters, and used them to monitor the status of the flight. When the height began to decrease, the flight computer determined that apogee has been reached.

The first stage parachute was released, which is a small drogue that stabilizes the rocket in flight as it descends. It was deployed by releasing pressurized CO₂ into the parachute bay. This snapped the shear pins holding the nosecone in place, popping it off the rocket. The action deployed the drogue in the process by blowing it out of the parachute bay. The second stage remained stowed in the parachute bay, restricted by a line.

In this design, the second stage deployment of the main parachute was meant to occur when either of the altimeters detects that 1500 AGL has been reached. At this point, pyrotechnic cutters would be activated to sever the line holding the parachute in inside the parachute bay. The main parachute would be pulled out by the drogue and inflate, slowing down the rocket for a descent at the correct speed.

For two years, the system did not work as intended. There were issues with the electrical connection, loading shock, and the performance of the pyrotechnic cutters. This culminated in the failure of the second main stage to deploy. The specific issues encountered in the past are beyond the scope of this report, but they provided useful information about the requirements of the new systems and led to the consideration of alternative design choices.

The first primary requirement of the system is that it must be safe. Operators should never be in danger from pyrotechnics when assembling or handling the rocket. The system

must obviously comply with the competition rules. Other broad primary requirements of the system are reliability and redundancy. It must reliably deploy both stages of the parachute system so that the rocket successfully lands in all circumstances. All critical components should be duplicated if possible, and in some places duplication is mandated by the rules. The final main requirement is that the system should be possible to develop and prepare for flight in one year, but as mentioned earlier this was extended to two years.

Secondary requirements for the system were also identified. The system needed to possess the ability to deploy stages quickly. Delaying the deployment time of either stage could introduce additional shock loading to the rocket. In the case of the main parachute, deploying its deployment too long could lead to there being insufficient space to slow down before ground impact. Cost was another parameter that we tried to minimize, as future funding amounts for the team are uncertain and a cheaper recovery system could allow money to be re-allocated to other subsystems.

The system also needed to be light, compact, and easy to assemble; weight savings could allow the overall rocket to be smaller and carry less fuel. That is easier to transport, assemble, and develop; less fuel also makes the system safer to work with. A system also needed to be easy to test and flexible in modification, so that we could effectively validate and tweak the system if necessary. Our last secondary requirement was a creative or unusual approach that would be interesting to develop and would benefit the team score at the competition.

2.1 Direction of initial development

The previous system was taken as a basis for the new system. However, many elements were put under scrutiny as if we were designing a brand new system. Several design decisions needed to be made, defaulting back to the old system if a advantageous design alternative could be found. This was also done to save development time, as it was one of the major constraints.

The structure of the flight computer was kept the same. Due to some specific rules about altitude logging requirements, a more advantageous alternative solution could not be identified. While adding a third redundant altimeter and the associated pyrotechnic device could have increased the reliability of the system further, this was not done for several reasons. Adding all of this would have increased the weight and size demands of the recovery electronics, and required a substantial re-design of the electrical system as a whole. Finally, the altimeters and their associated pyrotechnic cutters cost a non-trivial amount of money.

The CO2 mechanism of the primary deployment event was also retained. That mechanism had never failed during past competitions and thus proved to be very reliable. The two parts that were identified as having promising alternative design decision were the cutting devices, and parachute itself.

3.0 Cutter Background and Design Considerations

The first part to be put under substantial scrutiny was the line cutting device. The component used in the old system was a pyrotechnic cutter. When an electrical signal is applied to them, the pyrotechnic charge inside the cutter detonates, propelling a cutting rod into the line it is attached to. The direct failure of these cutters to sever the line was the major reason for the last failure of the system.

These current cutters had a number of downsides associated with them aside from the demonstrated unreliability. As mentioned previously, they are relatively expensive, and they are difficult to test and re-use. After firing, they must be disassembled and cleaned with acetone, then dried. Aside from being time-consuming, this makes it difficult to properly evaluate them through testing. The fact that they are a pyrotechnic device means that there are safety considerations when using them.

Several alternatives were conceived. The main ones were:

- Switching to CYPRESS cutters [4]; a higher quality type of pyrotechnic cutter used in human parachutes.
- A nichrome wire that melts through the nylon line.
- A custom pyrotechnic cutter developed by the team.
- A custom cast pyrogenic epoxy around the line that burns it when ignited.

The first alternative option with Cypress cutters had several upsides. Due to the fact that the Cypress cutter was human rated, it was extremely reliable. It's usual application is cutting a line to deploy the reserve parachute on a human skydiver, and it has allegedly never failed. It was also specifically designed to sever the type of line we were using.

However, its drawbacks were more substantial. Aside from being far more expensive, it was not re-usable. It retained the safety issues associated with pyrotechnics. Lastly, it possessed a unique connector which would have been difficult to interface with the electronics. Correspondence with suppliers also revealed that sourcing it at all would prove to be challenging as well.

The nichrome wire option was considered due to it being commonly used by other teams in our competition. Nichrome is a special alloy of nickel and chromium that is used for heating applications [5]. When electrical current passes through it, it heats up easily while maintaining favorable chemical and mechanical properties.

As the lines that our parachute uses are made of nylon, the heated nichrome would swiftly melt through the line, severing it. In this configuration, it was much cheaper and easier to use than the pyrotechnic cutters, and did not have as many safety concerns as it had no pyrotechnic elements. It was also easy to scale down for testing of smaller scale prototypes.

Reliability with the nichrome system seemed dubious, since a loose or frayed wire is possible, and could pose a huge risk to the functionality of the system. It was also slower to cut the rope. Finally, it would require a lot of current to heat up, which would again require substantial modifications to the electronics system. As it was a large departure from what the team had done previously and was not a novel system, the time required to develop it was also considered as a downside.

The custom pyrotechnic cutter was briefly explored, but it essentially possessed all of the downsides that the regular cutters did. The only advantages that could be gained were a reduced cost and higher availability, plus a greater flexibility of the system for modifications.

A novel and thus attractive strategy that was conceived is the use of a pyrogenic epoxy to burn through the lines. The selected material, RNX [6], could be cast right on the lines and ignited with an electronic match. The price of this would be exceedingly cheap. The materials for preparing the casts could be ordered in bulk, and the ignition of only a small amount was thought to be sufficient in order to cut the line. This system would also be very flexible.

The last default option was to retain the previous system with a few upgrades. An analysis of the failure revealed that the system could be made a lot more reliable by implementing better cleaning and testing procedures with the existing cutters. This method was the least time-consuming. Since it had already flown on the rocket, the flaws of the system were well-known and could be addressed, making this method reliable from a development point of view.

The decision that was made was to pursue the pyrogenic epoxy. This decision was made due a similarity to the current design and a large degree of flexibility and ease of testing. The novelty of the design was also factored into the decision.

3.1 Cutter Development

The first issue that arose when developing the cutters is that casting them on the line bound them tightly to that section of it. If they were cast on the control line and the one located

higher on the line would fail to detonate or detonated with a delay, the severed line could rip it off the rocket, possibly damaging the electronics in the process. Furthermore, the initial shock of reefed deployment could damage them right at the beginning. Thus, a two-ring release system was conceived that would de-couple the cutting devices from the line itself. Since this sort of system would be beneficial for almost any cutter system, it was pursued. A concept image can be seen in the figure below.

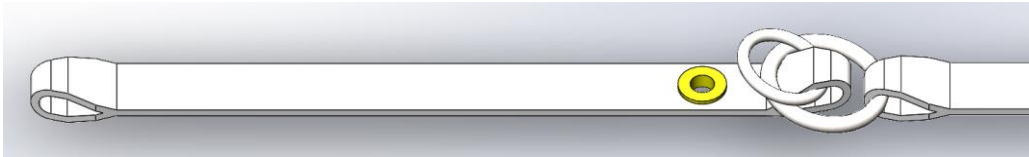


Figure 2: Two ring release

As testing began, another more pervasive issue was encountered. It took the epoxy several seconds to burn away and thus detach the line, which was unacceptably long. In that time, the rocket could travel over 500 feet and have an insufficient amount of space to slow down. This issue was still being worked on until the requirements of the design changed later on in the project. This will be discussed in a later section of this report.

4.0 Parachute Background and Design Considerations

For a more detailed description of the parachute system, some terms need to be defined. The parachute is constructed out of a number of fabric gores that are stitched together at their edges. This forms the canopy of the parachute, with the skirt around the bottom edge. A vent is left on top to let some air escape from the parachute. In between the gores at the skirt, shroud lines are attached that link to a single attachment point. From there, a shock cord extension line extends down to a swivel and the load, which in this case is the rocket. A diagram of a similar system is shown below.

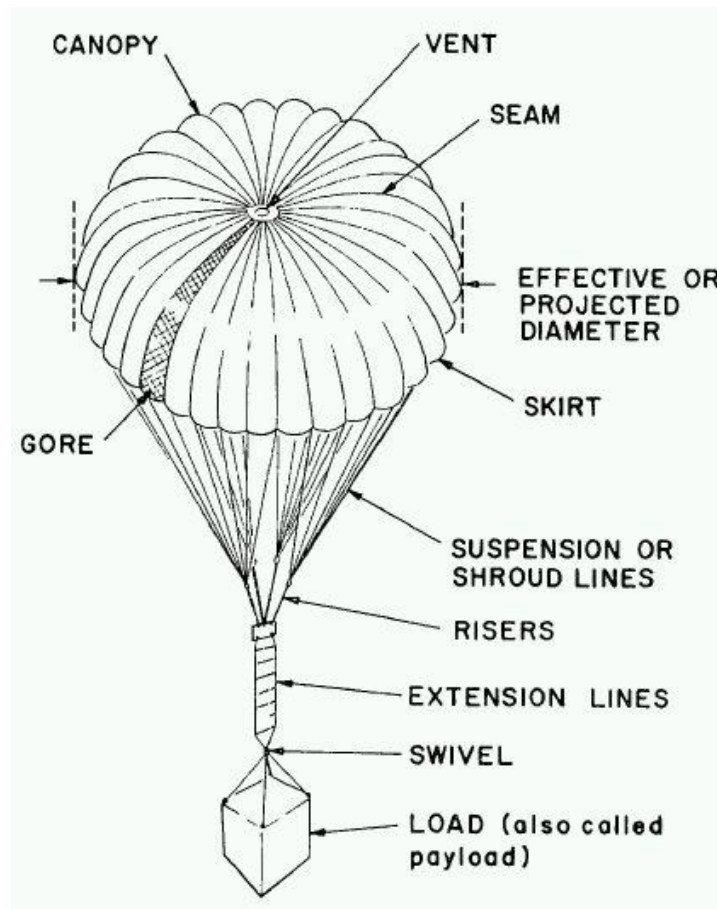


Figure 3: Diagram of relevant parachute parts [7]

The parachutes that the team used had worked reliably in the past. The two stages of the parachutes were accomplished through the use of a drogue and main parachutes, as described previously. The parachutes utilized a semi-elliptical geometry with a 1:0.707 ratio, which maximizes the performance of the system [6]. Another reason for its use was the

existence of convenient manufacturing stencils generators for this design. It was decided that this shape would be retained for the next cycle.

An interesting alternate system was conceived for the upcoming competition cycle. Instead of using a separate drogue and pilot parachute, we considered using a single parachute reefed system. Reefing is a term borrowed from sailing, where reefing a sail is a term referring to decreasing its surface area [8]. Similarly, parachute reefing is the process of restricting the parachute from opening fully.

There would have been several advantages to this system. Apart from being a somewhat creative approach, it would also decrease the shock loading of the system during the disreefing process. Having only one parachute instead of two would also give the opportunity for weight savings.

The only major downside to this would be the requirement for a small pilot chute that would pull the reefed main parachute out of the parachute bag. That meant that the true weight savings were not as substantial, and some testing would need to be done in ensuring that it worked reliably.

4.1 Physical & testing requirements of parachute system

The requirements for the two stages of the parachute are to be able to descend the rocket at roughly 112 and 30-20 feet per second at altitudes of 30 000 and 1500 feet AGL, respectively. However, this needed to be quantified in specific low-level engineering requirements such as size and shape. Since there was a large window for both of these values, the parachute did not need to be extremely accurate, but verifying it was necessary. To do this, an additional requirement was put forth in the form of a comprehensive testing platform for the parachute.

The parachute is governed by the drag equation [9], noted as equation (1). It applies to all objects travelling in a fluid such as air.

$$F_D = q \cdot S \cdot C_D \quad (1)$$

S is the surface area of the object, C_D is the dimensionless coefficient of drag, and q is the dynamic pressure, which can further be calculated from the velocity of the object and the density of the fluid, in this case the air density. The equation for this is:

$$q = 1/2 \cdot \rho \cdot v^2 \quad (2)$$

Assuming that the sum of forces in flight are the forces due to drag and gravity, the critical equation for the descent rate can be written as shown in equation (3). W represents the weight of the system.

$$v_d = \sqrt{\frac{2W}{C_d S \rho}} \quad (3)$$

A complicating fact to this is that the surface area is somewhat hard to define. The nominal surface area S_o is usually used for calculations, which considers the total surface area of the fabric. Consequently, the nominal diameter D_o is a term defined as the diameter of the total fabric area if it was laid flat in a circle. As would also become apparent during testing, the inflated parachute diameter can also differ substantially from the diameter that the parachute is constructed to.

The surface area S term is thus confusing to use overall. As such, the combined term of $(C_d S)_o$, the nominal drag area, is the relevant quantity that represents the construction of the parachute. Calculations for cost, weight, and volume efficiency are derived from this term. When C_d is used, it is usually used in conjunction with the nominal diameter D_o .

While it can be approximated from known values, the drag area of any constructed parachute must ultimately be determined, tuned, and confirmed by testing. An added challenge at this point was the lack of existing parachute data for the semi-elliptical geometry we were using. The best that could be determined is that the C_d values would be somewhere close to the range given by polyconical parachutes, which is given as 0.80 to 0.96 [10].

A reefed parachute specifically possesses another noteworthy parameter, $(C_d S)_R$. This is the reefed drag area of the parachute. From this, the ϵ value of the reefing can be defined as:

$$\epsilon = \frac{(C_d S)_o}{(C_d S)_R} \quad (4)$$

Which is the ‘reefing ratio’ of the parachute. Given the targeted drag areas for both stages as calculated previously, this would be roughly half for the current system. In order to

confirm that these values were correctly designed, a testing solution was desired. It needed to not only satisfy the short-term demands of the project, but be usable in future parachute design cycles as well.

Testing the parachute was not rigorously done on the team prior to this year. In the past, the parachute was flown in the wind with a handheld anemometer, and from that a rough coefficient of drag was determined. The surface area of the parachute was determined using the constructed diameter, and so a $(C_D S)_o$ was computed. Several more rigorous approaches were considered.

The obvious solution that is usually taken in this field of research is to use a classical wind tunnel. It is a chamber usually used for airplane model testing, where fans drive air to generate wind the chamber to simulate flight conditions. Parachute models can be tested rigorously in such tunnels across a variety of conditions. However, logistical difficulties for doing this were encountered.

The university wind tunnel was only a meter in diameter, which was considered too small to test the parachute or a representative scale model of it. The full parachute's constructed diameter was almost 4 meters, and recommended blockage ratios for wind tunnels are typically less than 5% [11]. This would mean that the small scale parachute would need to be almost ten times smaller than the real thing, which was not considered to be representative. The other compounding issues is that a cutting device for this setup would also be difficult to test, and specific small lines would need to be purchased just to run the test. Manufacturing such a small parachute accurately would also start to become difficult for our team's manufacturing capabilities.

Large wind tunnels were available outside the university, but they were prohibitively expensive to use, had no gaps in their schedule, or did not permit the kind of testing we wanted.

Other options such as airplane tug tests, mentioned extensively in Knacke's manual [10], were deemed too difficult to organize and too expensive. However, a modification could be made that would put this within the capabilities of the design team. This option would be a car tug test instead. In order to successfully tow a parachute of arbitrary size behind a vehicle, a special rig needed to be designed, which would be time consuming. For it to work, a long, straight section of road that was required, but this could be provided by the Region of Waterloo's emergency services test track. This approach was selected nonetheless due to being the best method within the organizational and engineering capabilities of the design team.

The design of this testing platform became a smaller engineering problem by itself. It was desired that the testing setup could test parachutes of any size up to the full scale parachute and possibly beyond. This also needed to be done in a safe manner, and had to be possible to deploy on the test track we were provided with. The simplest, lightest, most representative way to do this which has the lowest risk of damaging the parachute would be selected.

The most simple way to do this would be to simply strap the parachute to the floor of a pickup truck, possibly with an operator holding it, and then accelerating. The safety of this method is dubious at best if the presence of an operator in the back of truck is necessary, and larger parachutes would become difficult to test. A large issue with testing the parachute in this manner is wake effects from the truck. Wake effects can cause breathing of the parachute [10], and make measuring the wind hitting the parachute difficult, which is necessary for performing drag area calculations as discussed previously. The risk of the parachute dragging on the ground and getting damaged also become elevated in this configuration.

The testing setup suggested by Richard Nakka [6] is a pole extending out of the side of the car. This did have the advantage of mitigating some of the wake effects, and was safer both for personnel and the parachute. However, the size that this configuration could test only limited sizes of parachutes due to the asymmetrical loading on the vehicle. If higher drag forces become generated, a substantial turning moment will also be generated the car will become difficult to control, which is a safety concern. There were also concerns about maneuvering with such a setup around the test track.



Figure 4: Car testing setup as proposed by Richard Nakka [6]

A variation on this design was considered. The pole would be mounted vertically to access the air flow above the wake of the truck. The forces generated would be above and in line the vehicle, which means that the generated moment would result in a pitching of the

vehicle. Due to the weight of the test vehicle, a pickup truck, and the expected winds, this scenario was ruled out. Maneuvering with such a setup was also considered to be easier since the parachute was above the vehicle, and there are no bridges or overpasses on the test track. One downside of this would be that there would be added difficulties in mounting the equipment to the top of this pole, but this comes with a reduced risk of the parachute dragging on the ground.

To increase rigidity of the system, straps were added to the top of the pole. After a force analysis, it was determined that the system was safe to use, and the rig was manufactured. It connected to a hitch connector of the pickup truck. A small release mechanism allowing disreefing of the parachute was added near the top. Mounting points for a phone acting as a camera, anemometer, and load cell were also added to the top. The camera would record the windspeed on the anemometer at each instant. The load cell would send data into the car through a cable, where it was connected to custom data acquisition electronics and a computer. With the load and wind speed measured, and all other values as known constants, drag areas of the parachute could be determined.

An additional step was also taken to lessen the loads on the rig and ease operations. A roughly half-size parachute was constructed that would be easier to rig and transport. However, this strategy would lead to unforeseen difficulties that became apparent only after testing commenced.

4.2 Detailed design of the reefing parachute & preparation of first prototype

Several reefing configurations were considered, drawing from the few limited papers that could be found on the subject. Four initial designs were selected for analysis. They were analyzed for reliability. Ease of implementation was also considered.

Skirt reefing, as used in the Apollo program [10], would have involved putting cutters around the skirt of the parachutes. The rigging for this setup would be exceedingly simple, with a single circular loop of line around the skirt of the parachute. In the Apollo system, a pyrotechnic fuse time delay was used in the cutters positioned around the loop, which meant they required no additional attention after parachute deployment. Aside from moderate parachute modifications, this system looked promising.

However, the rules of the competition imposed a complication to this setup. The electronics controlling the pyrotechnics needed to be redundant, and the deployment event

needed to be altitude-controlled, not time-delayed. One way to address it would be to mount substantial portion of the electronics needed to be mounted on the parachute skirt, which was not feasible. The other more realistic solution was to run wires from the rocket up to the skirt, but this was still problematic because of the substantial weight demands and reliability concerns. Radio-activated cutters could be better, albeit with a non-trivial amount of electrical modifications and testing required. Furthermore, the reliability and safety issues associated with radio-controlled cutter were even greater. Due to the lack of viable design strategies, this method was rejected from further consideration.

The other method with simple rigging was vent reefing, where the parachute is pulled in from the vent to constrict the parachute into an odd shape. This was considered to be very easy to rig and integrate into the current system. There was a significant difficulty in even estimating the inflated diameter of this configuration. At the time, this was seen as a major downside. Nonetheless, it was kept in consideration.

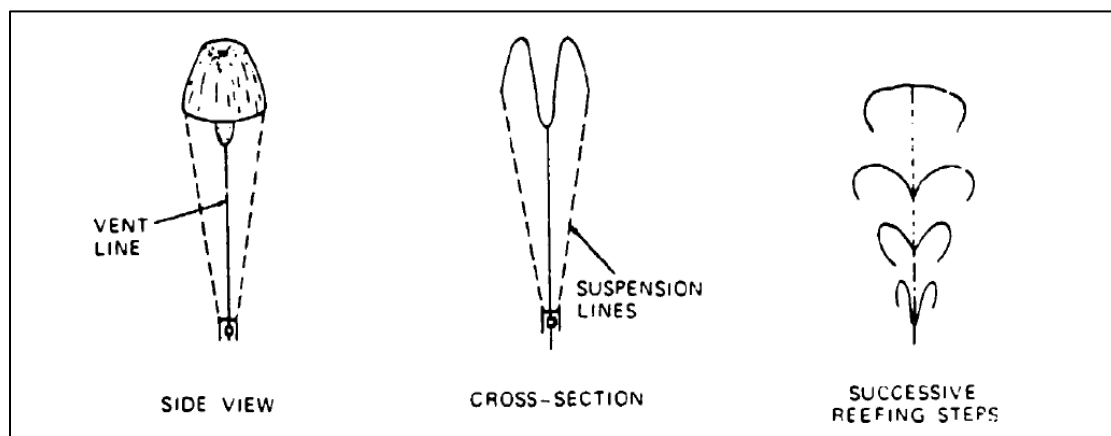


Figure 5: Vent reefing diagram [10]

The last method considered was control line reefing. This was similar to skirt reefing, but the line around the skirt is sized for the full inflated size, and then constricted by being pulled down by another control line. This control line can be released from the rocket to disreef the parachute, which meant that this was easy to integrate into the recovery system. The exact rigging could then be varied further to a dual line and single line configuration, which can be seen in figures 6 and 7, respectively. There were several papers describing these various configurations, which led more credit to the method.

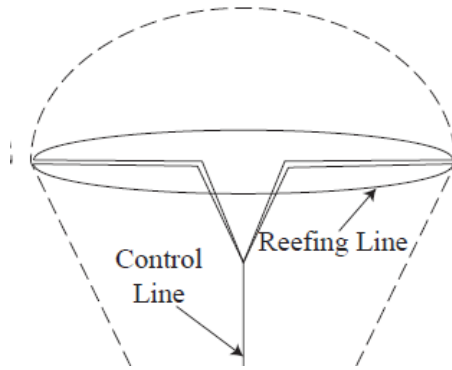


Figure 6: Dual line skirt reefing rigging configuration [12].

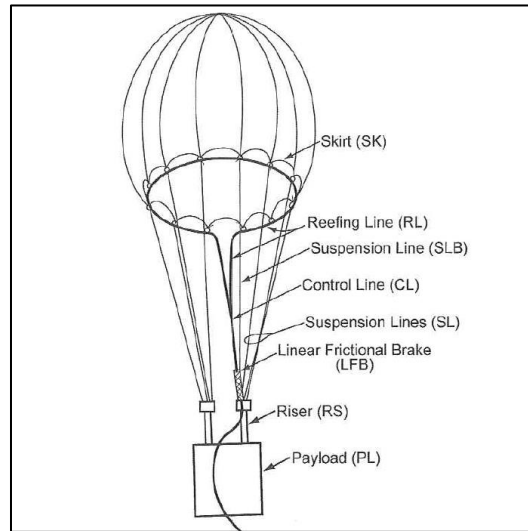


Figure 7: Single line skirt reefing rigging configuration [13].

The method selected for testing was control line reefing. The ease of integration into the existing system and simple rigging was the same for both systems. However, the larger body of resources and rigging options that was found for control line reefing, which led to its selection. Another thing which was taken into account was the simpler restricted geometry of the control line reefed parachute. At this time, it was still thought that geometries, line lengths, and drag areas could be estimated with some degree of accuracy using theoretical calculations. This would be proven incorrect as soon as testing commenced.

5.0 Parachute Testing

The plan for the first series of tests was simple. The small scale parachute would be tested to a targeted 25, 50, and 75 percent of it's drag area. This would be done several skirt reefing configurations, with several runs also disreefing partway through. Based on the results of the test, the best method of skirt reefing could be determined, and the drag area estimates compared with real values.

The small scale parachute to be used for the test was augmented with small metal rings at each gore. This would allow the skirt line to prevent the full opening of the parachute. Metal rings were used to resist all tensile forces from both the parachute and skirt line, and to also minimize friction between the skirt line.

Testing commenced at the test track in mid-December of 2020, in snowy conditions. It was hoped that the lines could be simply sized to the constructed geometry with only minimal error. The drag areas could be verified through testing, and applied to the larger scale parachute for the flight.

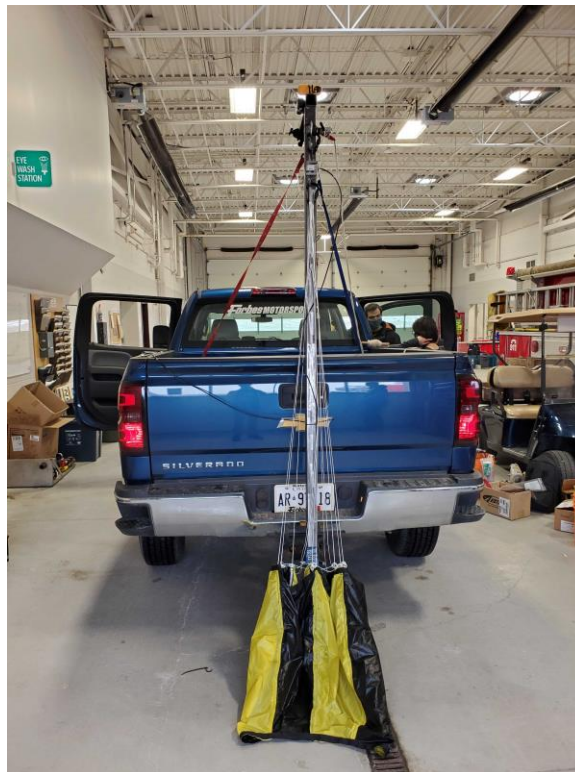


Figure 8: Test setup for first truck test

Initial lines were rigged to the dual skirt line configuration. The rigging process was slow and testing fell behind. When two runs were done using the truck test setup, and several

problems immediately became evident. The inflated diameter turned out to be much smaller than the constructed diameter. The reefing lines that had been sized to constructed dimensions were too large and almost entirely useless.

Exacerbating this fact was the low gore count of the parachute. This meant that a very substantial amount of the parachute bulged beyond the reefing rings, as can be seen in the figure below. These two facts combined meant that incredibly short lines were required to constrict it at all, and re-calculations were hastily done at the test site. Targeting any specific drag areas was dropped as a goal for the test day. Disreefing tests and the evaluation of the optimal rigging approach became the primary objective.

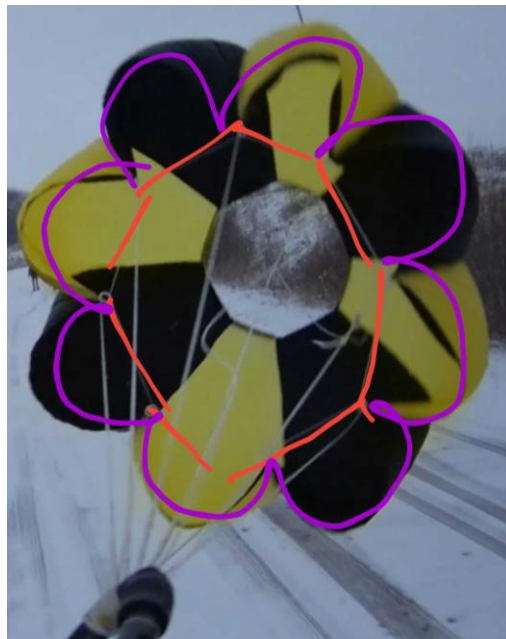


Figure 9: Bulging of parachute skirt (in purple), and the skirt line (in red). [14]

The first approach was deemed to be overly complex in terms of rigging, and tangled once during the tests. It also required a large length of line. For these reasons, the single side-line approach was decided on as the single design moving forward.

With this last configuration, several successful disreefing tests using a roughly sized control line were executed. A figure stitched together from operational footage can be seen below.

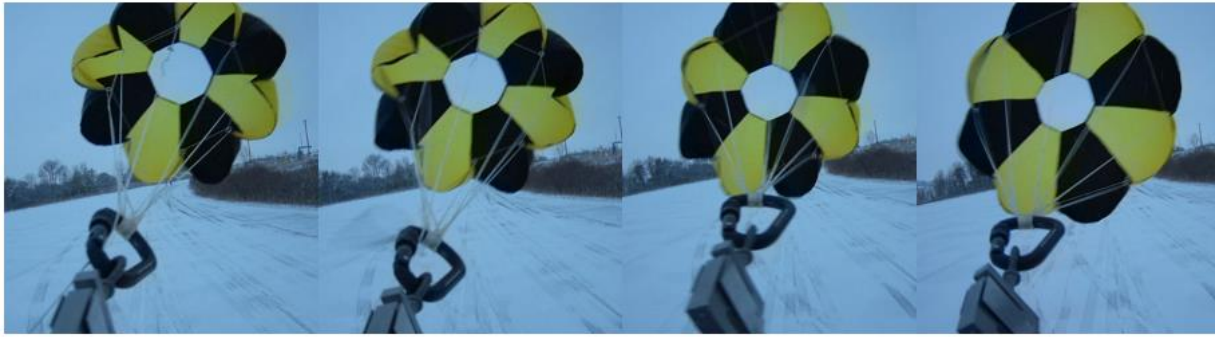


Figure 10: Small scale parachute disreefing sequence

The data processing after the test was very difficult, and most of it was never analyzed. It did become clear that there was extreme variation and noise in our data. What should have been an exact quadratic relationship between windspeed and force scattered more and more with increased windspeed. The drag area over time graphs also contained substantial variation. It was concluded at this point that future testing would need to be much more thorough to average out these variations.

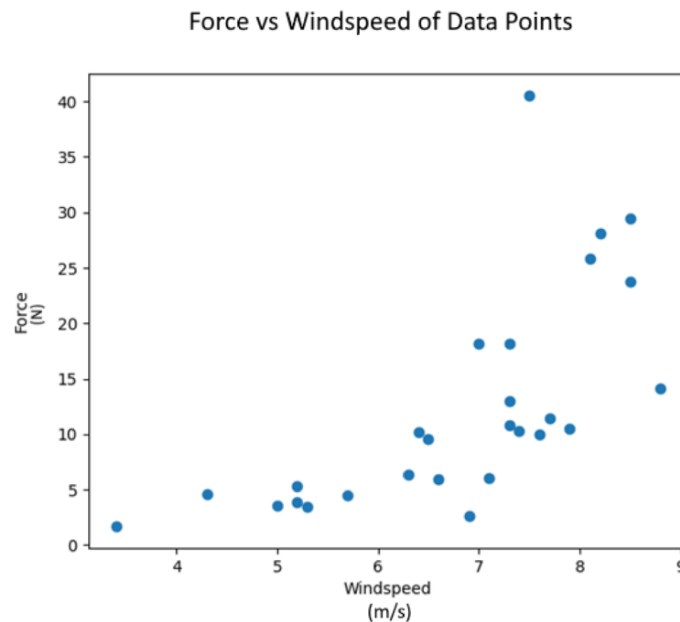


Figure 11: A plot of the Force-Windspeed relation of all collected data points, which should be strictly quadratic.

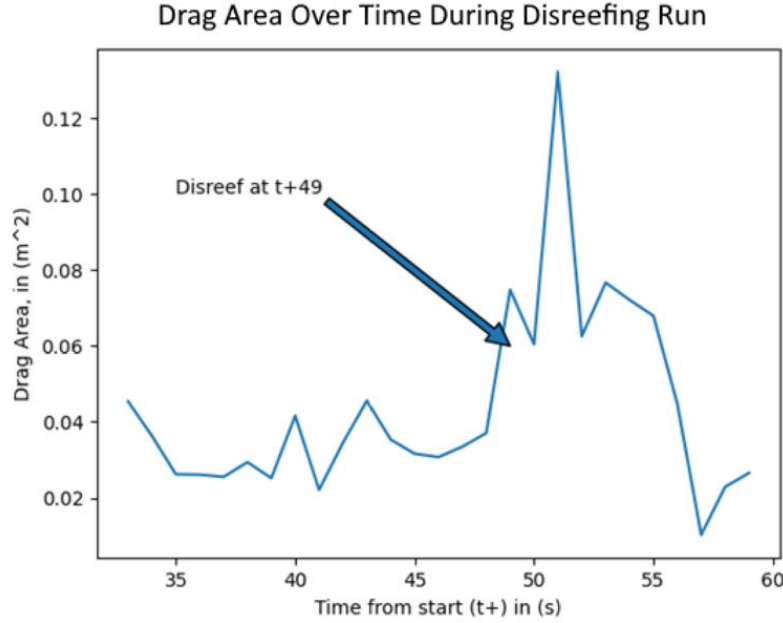


Figure 12: The variation in drag area over the course of a disreefing run

5.1 Re-evaluation of experimental methods

After this experience, further research on the physics of the system was done. It became obvious that the inflated diameter was significantly smaller, and an attempt was made to estimate it using existing research. For this purpose, Knacke's *Parachute Recovery Systems Design Manual* was used extensively. The only estimate for an inflated diameter that was found was from Knacke [10], given as a $0.72 D_p/D_o$ ratio, where D_p is the inflated and D_o the nominal diameter. This value was listed for tri-conical parachutes with specific fabric properties, so it would not be accurate enough to apply with confidence.

The other conclusion was that there would be no fast and accurate way to approximate the drag area generated from a specific arrangement of lines. A value that could be directly controlled is referred to as Tau. Tau was a value that encompassed the length of the reefing radius and other parachute constants, defined as:

$$\tau = \frac{D_R}{D_o} \quad (5)$$

Knacke mentioned that a tau-epsilon curve needed to be generated for a new parachute. The generation of such a curve would allow the reefing of the parachute to any drag area ratio. An example is shown in the figure below.

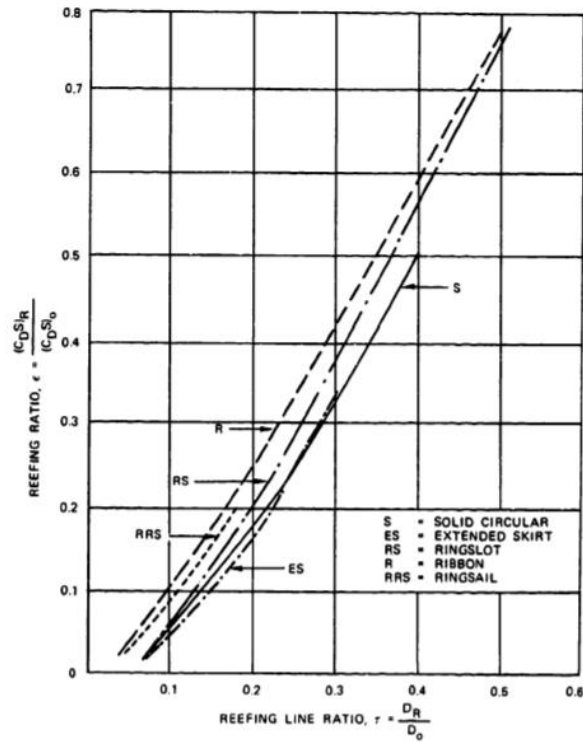


Figure 13: An example of a tau-epsilon curve from Knacke's manual [10]

The issue of the bulging fabric was also researched. Some sort of correction factor needed to be applied. Knacke recommended a solution using the following formula:

$$D_R = D_0 \cdot \tau \cdot C \quad (6)$$

Where D_R is the reefed diameter, and C is the correction factor. This correction factor varies with the number of gores. A graph was provided by Knacke, as seen below.

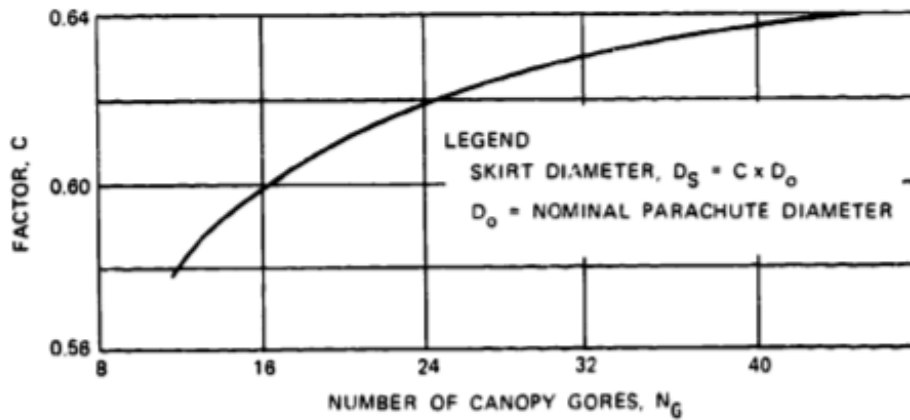


Figure 14: The correction factor for the number of gores in a parachute [10]

The new testing plan called for a more broad test range with a τ range 0.2 to 0.8 . A number of small line segments were manufactured so that the line could be modified easily at the test site. These lines, called 'tau lines', could increase the tau value of the system by 0.1 when added. Knacke's correction factor would also need to be used. However, since the correction values are not given for eight gores, a decision needed to be made for further testing. The options were:

- Continue using the small scale parachute and extrapolate the line to come up with a correction factor.
- Modify the small scale parachute by adding rings in the middle of each gore, effectively increasing the gore count to 16.
- Create a brand new small scale 12 gore parachute, the same value as our already-fabricated flight parachute.
- Use the flight parachute for testing, modified with metal rings for reefing.

The first method was deemed to be incredibly unreliable, and was not considered after some deliberation. The second method would take some moderate fabrication work, but provide us with a more accurate correction factor. Moreover, it would allow us to retain all the convenience advantages of a small-scale model, and was thus seriously considered. The third method was deemed to take too much time, and was discarded. The use of the flight parachute was also a very attractive. There would be no chance of a scaling issue between the small and full scale, and little work would be required to modify it. This last strategy was chosen, despite the anticipated challenges in rigging and handling it. These challenges would go on to impede the system as testing continued.

Another few modifications we made at this time. The height of the test rig would be extended using a custom extension adaptor, such that there would be less effect from the wake on the system. The other was conceived to make the rigging even more simple, and measurements more straightforward. Instead of having a separate skirt and control line, a single hybrid line could serve both functions when tied off at the starting ring. This would simplify the rigging into a single line. While drag issues could have been problematic, it could be tested with relative ease during the next test. A diagram of this setup is shown below.

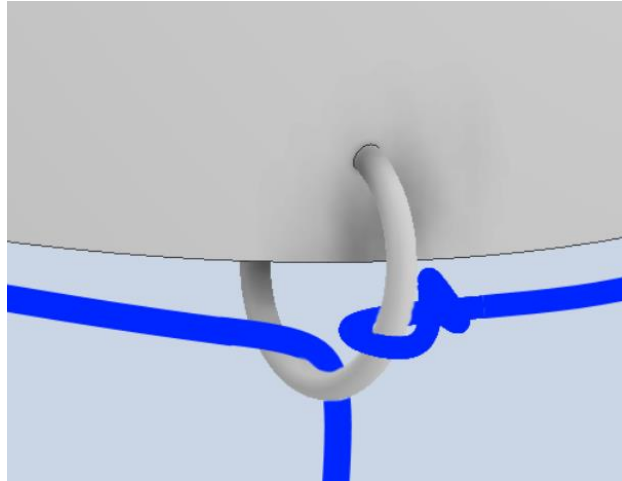


Figure 15: Illustration of hybrid line rigging.

5.2 Second test cycle

This test, more issues were encountered immediately upon arriving at the test site. Due to issues with our load cell, accurate load data could not be determined. Furthermore, inaccurate dimensioning of the large scale parachute once again introduced massive problems to the pre-prepared rigging. Due to the high ambient wind on the day, handling the full-scale parachute became incredibly difficult and thus test turnaround was slow. Oftentimes, it took several people to lower and wrestle the parachute closed. In summary, limited amounts of quantitative data could be determined.



Figure 16: Test setup for second truck test [14]

Some qualitative data could be determined, however. The hybrid line method as described earlier proved to have no issues and was simple to rig even on the full scale parachute. Disreefing tests were done with it, as can be seen in the figure below, and it performed well. Based on these findings, it superseded the previous iteration of the side line rigging and was used from that point on.



Figure 17: Disreefing sequence of the full scale parachute, using a hybrid line.

The other key findings was the inconsistent inflation and oscillations of the parachute when significantly reefed. Sometimes, instead of inflating to a long shape, the parachute closed into a line. This was concerning, but since substantial drag force was still generated by this shape it was deemed acceptable. Similar effects were later discovered in recovery footage from SpaceX, so this was not a significant cause for concern.



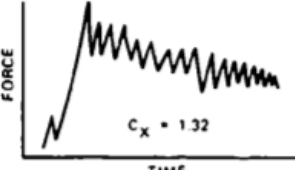
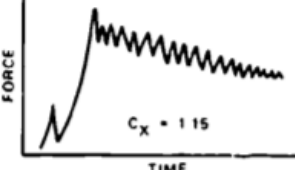
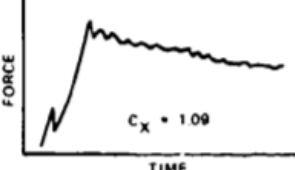
Figure 18: Odd shape of parachute during a failure to inflate. [14]



Figure 19: Similar odd shape found on SpaceX footage. [14]

The oscillations were also not a significant cause for concern, as some oscillation is expected from forebody wake, especially from. An example of this phenomenon can be seen in table 1 below. Since the tolerance of the rocket to forces and oscillations is very high anyway, and an average descent speed will be reached, it would not affect the rest of the design significantly. It was also unclear if the wake would even be generated in substantial amounts when mounted to the rocket and not the truck. To imitate this, the strap holding back the parachute was lengthened in preparation for the next test.

Table 1: Effects of forebody wake on drag oscillation. [10]

FORCE-TIME TRACE	VEHICLE CONFIGURATION	DESIGNATION	VEHICLE DIAMETER, FT
 FORCE TIME $C_X = 1.32$		BOILERPLATE COMMAND MODULE (B/P CM)	11.9
 FORCE TIME $C_X = 1.15$		PARACHUTE TEST VEHICLE (PTV)	5.8
 FORCE TIME $C_X = 1.09$		INSTRUMENTED CYLINDRICAL TEST VEHICLE (ICTV)	3.0

6.0 Change of requirements

After the second test, a culmination of substantial developments changed the course of the project. It was late March 2021. All throughout the project, timelines had continued to lapse due to the evolving coronavirus situation. However, it seemed like a launch event could finally take place in October of that year, and the system had to be flight-worthy by then. However, this was now for a different competition than Spaceport America, so the rules there would not be as strict. This gave more flexibility in the design.

Administrative work by the team also played its part. A batch of brand new pyrotechnic cutters was procured for free from a sponsor. Moreover, talks with a helicopter charter company that had been ongoing for some time had been finalized. The team now had the capability to perform a high-altitude drop test of the recovery system.

As a result of these developments, the pyrogenic epoxy was no longer necessary and was dropped from the design. Due to the looser rules of the competition, the drop test could also alleviate the testing issues being encountered on the ground. Carrying out that test would come with its own challenges, but this was a significantly more productive and straightforward way to proceed. A full-scale validation of the entire system would be tremendously helpful in catching any design flaws. Aside from logistics, the only major new restriction it introduced upon the system was the substantially higher risk of drift off of the drop area due to the small size available to the charter company.

A new overarching plan for the system was created. On the ground, all that was now required was to find a line length for the parachute that creates an acceptably low drag area; a drag area small enough to safely carry out the drop test. Once the test is carried out, a very accurate descent rate would be known for the parachute. After that, the length could be simply tweaked to fit within the looser guidelines of the new competition, should that even prove to be necessary.

Therefore, the new design goal of the project was to successfully carry out this helicopter drop test. After that, the system would be deemed ready to fly on the rocket during the competition. With this shift of priorities considered, a third truck test was carried out.

7.0 Testing under revised requirements

For the third truck test cycle, technical issues were not a concern. The only problems encountered were more discrepancies in the rigging lengths, and the length of the tau lines. Originally, they had not been made for the hybrid line configuration, and were now too short to reduce the tau value by 0.1. However, since the goal was now to find an acceptably small drag area, the test plan was augmented.

The parachute was simply reefed by hand to the point where it would no longer inflate, and after that point tau lines were incrementally added back in. Up until 4 lines, consistent inflation could not be achieved. Wake problems were still prevalent in spite of the extended strap. Aside from that, testing was carried out successfully, but only a limited number of runs could be performed on the day. An example of a processed run can be seen in the figure below.

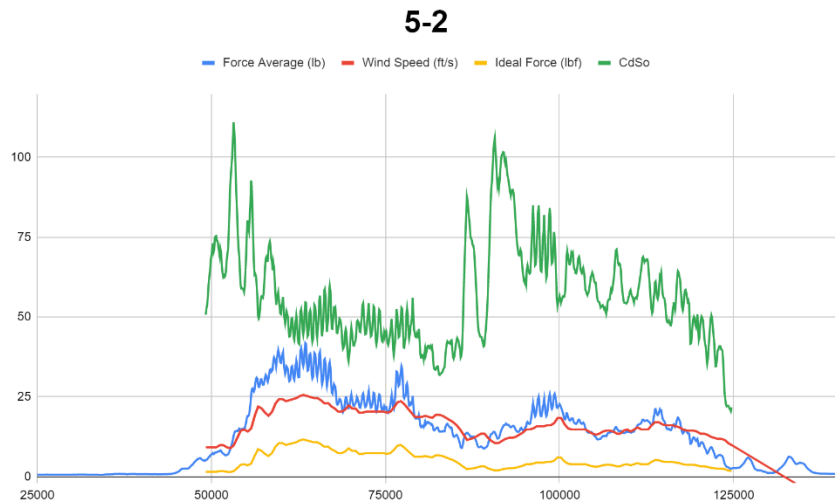


Figure 20: All parachute parameters plotted over the course of run '5-2'

When data was analyzed in the weeks following the test, there were a few times when there seemed to be a discrepancy between the wind speed and generated force that was not accounted for by the shape or inflation of the parachute. While this could not be confirmed for sure, the source of error was likely due to the inclusion of the height extension and longer strap. Instead of mitigating the effects of the wake, they had exacerbated them. By increasing the height, the instrumentation was less affected by the wake. The parachute with the longer strap, on the other hand, could now dip into the wake. During these periods of time, the anemometer readings were disconnected from the actual wind hitting the parachute. As no correction could be applied, these periods of erroneous readings had to be tolerated as a source of error.



Figure 21: Visualization of wake from pickup truck [15]

Nonetheless, a rough tau-epsilon curve was formed from the results, but did not look impressive. The variation was still very great between runs, and an insufficient range of values was generated to obtain a proper curve across the whole range of values. It was clear that testing at an even greater quantity would need to be completed if a curve of higher quality was to be generated. However, since that was not required anymore, a shortened hybrid line with 4 tau lines added was set as the drop test configuration as the shortest consistently inflated setup.

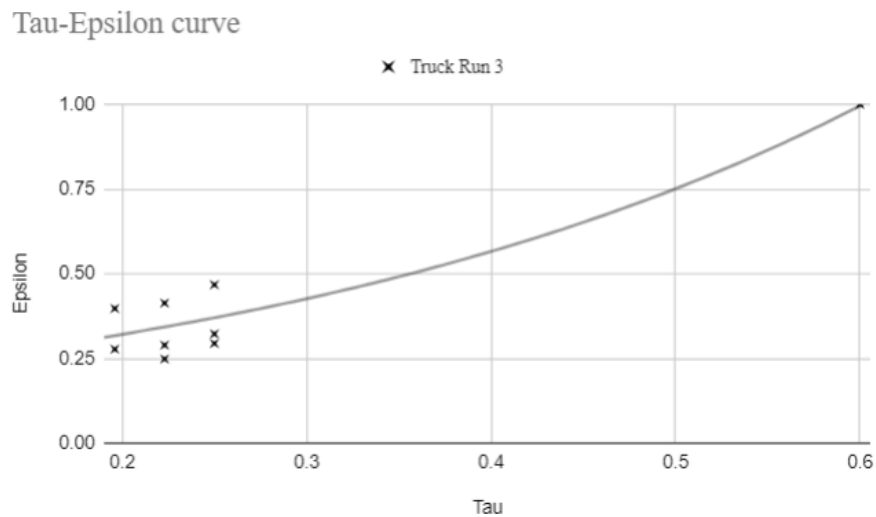


Figure 22: Tau-epsilon curve generated from the results of truck test 3

7.1 Pilot parachute design goals and development

The last thing remaining to be done for the parachute at that time was the attachment of the pilot parachute. As discussed previously, this is a small parachute which extracts the main parachute from its bag. The bag is necessary so that the lines of the main parachute do not get tangled in flight. For the pilot parachute used by the team, a mesh would be used in place of line to significantly reduce the risk of it tangling.

The recommendation is that pilot chute extraction force ought to be at least four times or more than the weight of the parachute to be removed from the bag during deployment. A pilot parachute was sized roughly to these parameters. Due to an increasingly tight timeline at that time in the project, only one prototype was made using existing parachute manufacturing techniques, and multiple designs were not considered. Before flight, it did need to be verified.

This was accomplished by performing a drop tower test on the system. The system was packed in the correct configuration. The main parachute was tied up to prevent it from interfering with the pilot chute and to ease retrieval of the system. A collage of the test footage can be seen below. The pilot chute successfully pulled out the main parachute in under 60 feet of travel distance, which was the height of the drop tower.

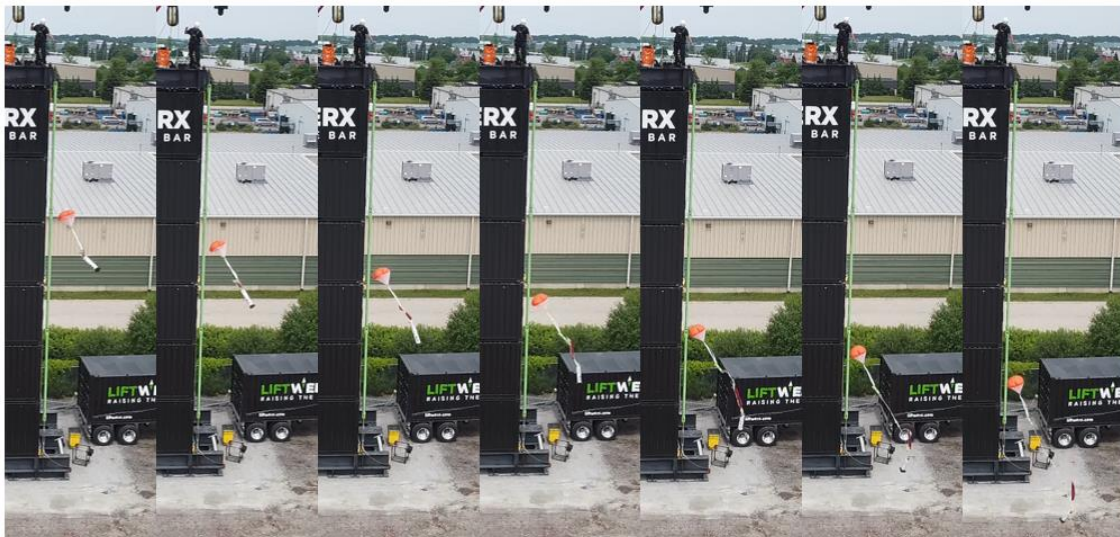


Figure 23: Successful deployment of main parachute from its bag using a pilot chute during drop tower test.

7.2 Helicopter drop test preparation

The test would occur on a small commercial helicopter, which would take off from a helipad near the Kitchener-Waterloo airport and proceed to a designated drop zone. The size of the craft meant that two operators could sit in the back seats with a test article, and drop when necessary. There was one more person who could sit next to the pilot and communicate with the ground.



Figure 24: Members of Waterloo Rocketry performing an initial inspection of the drop test helicopter. [14]

To execute the drop test successfully, the test article was designed. This device would be a replica of the top portion of the rocket, containing the flight computer and parachutes. To simulate the remaining mass of the rocket, a 80 pound deadweight was machined and attached to the bottom of this article. While the electronics inside were modified extensively for the specifics of the test, that is beyond the scope of this report.

The biggest design problem that needed to be solved was how to safely and reliably deploy the test article. It needed to both clear the skids of the parachute, and avoid getting caught in the helicopter blades. To accomplish this, a separate deployment article was thought of. A slide of sorts that sits on the skids could guide the test article past the skids and prevent it from deploying until it safely below the helicopter. Since it could interfere with the skids, it would need to be set in place once in the air. When the time to drop would come, the operators would lower the slide from their laps onto one of the skids. Two alternative designs could then be implemented.

In one design, they would then unstrap the test article also sitting on their lap, and insert it into the deployment article, allowing it to drop thereafter. In the other design, the deployment article would already contain the test article inside it, and would be released via a trap door. However, making a reliable trap door mechanism would be an engineering problem by itself, which was something that was not desirable. It is for this reason that the first design was selected as the primary one, but the other was still kept open in the event that the helicopter company would have a preference. Since they were carrying out the test, the team deferred to them on all critical decisions.

A protocol for the test was written up and approved by the helicopter company. The primary design for the deployment article was also approved by them at that time. That effectively concluded the project from a development point of view. The completion of the revised design goals would be confirmed by a successful test, for which all preparations were complete. The team was put on standby for flight the following weekend, weather pending. This was in mid-July, 2021.

7.3 End of testing

Days before the test was scheduled to go ahead, Transport Canada intervened and demanded that an additional permit be filed. Because of the incident that occurred earlier in the year with another student team testing parachutes, they had been notified. The team waited on standby for several weeks before Transport Canada denied the permit, cancelling the test.

At the same time, summer was coming to an end no details forthcoming about the October event. Setbacks beyond the scope of this report had occurred in other sub-teams, and a fourth pandemic wave was being forecasted for the fall. Under these conditions, the team made the decision to withdraw from the launch event, even if details about it would surface. Effectively, both the means and the reason for the recovery project in this competition cycle had thus been removed.

8.0 Conclusions

Even at the time of writing, the project's results remain inconclusive. All design objectives were met from a theoretical point of view, yet they could not be practically tested due to regulatory issues. At the same time, there was no longer any urgent purpose to pursue testing via another route.

The only design conclusions that were reached were basic evaluations of the cutter and parachute designs. The effectiveness of the single hybrid line as a reefing method was demonstrated. The delayed and problematic nature of the pyrogenic epoxy was also found. Unfortunately, the ultimate reliability of these components within the larger system could not be determined.

8.1 Next Steps and Recommendations

As this report is being compiled in the second half of September 2021, some work continues, and the progress made will carry over into future design cycles. The recommended Talks with Transport Canada are ongoing to determine what test conditions would be acceptable. The truck test experimental platform is being upgraded further for more accurate and extensive testing.

As such, the main recommendation for future development of the system is the creation of a more robust testing platform. With this, some of the logistical issues encountered could be avoided. This could be accomplished by further improving the truck tests system, such as through the procurement of digital anemometers, which would allow faster data processing and a consequently higher volume of data collection. From the administrative side of things, a proper arrangement with Transport Canada should be sought.

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